

Ioan-Daniel Craciun

PhD candidate in scientific machine learning, Technical University of Munich

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EDUCATION

2025–	Doctoral Candidate (Dr. rer. nat.) , Technical University of Munich Chair for Physics-Enhanced Machine Learning. Supervisors: Prof. Dr. Felix Dietrich and PD Dr.-Ing. habil. Stefan Adami. Started with surrogates for laser powder bed fusion (LPBF) melt-pool dynamics trained on large-scale Smoothed Particle Hydrodynamics (SPH) simulations; now working on latent-space, random feature and structure-preserving (Lagrangian, Hamiltonian) models with uncertainty quantification, aimed at reliable digital twins.	
2022–2025	M.Sc. Informatics , Technical University of Munich Focus on Machine Learning and Analytics. Thesis: <i>Molecular Conformation Sampling with Diffusion Models</i> . Combined diffusion models with an attention-based graph neural network that preserves inter-molecular distances, replacing force-field and repulsion-based corrections with a differentiable component.	Grade 1.5
2018–2022	B.Sc. Informatics , Technical University of Munich Application area Engineering. Thesis (grade 1.0): <i>Development and Control of a Contractible Hip for a Mouse Robot</i> . Designed and 3D-printed a hip system for a robotic rat that improves stability and enables narrower passages and new turning patterns.	Grade 1.8
2014–2018	High School , Mircea cel Bătrân National College, Constanța, Romania Mathematics and Informatics profile.	Baccalaureate 9.93/10

RESEARCH INTERESTS

Methods	Structure-preserving machine learning (Lagrangian and Hamiltonian mechanics); latent-space dynamics with operator- and measure-theoretic methods; random feature models; generative models (VAEs, diffusion models); Gaussian processes, active learning, level-set estimation and Bayesian optimisation; uncertainty quantification.
Foundations	Stochastic processes, Bayesian inference and MCMC, sensitivity analysis, experimental design.
Applications	Digital twins; additive manufacturing and fluid dynamics; benchmark datasets and reproducible research infrastructure.

CURRENT RESEARCH PROJECTS

2025–	Simulation Data Pipeline and Melt-Pool Dataset paper under review End-to-end, simulator-agnostic pipeline for generating, orchestrating and curating large SPH simulation campaigns on HPC infrastructure (SLURM scheduling, containerised execution, automated post-processing and labelling). To be released as an openly licensed melt-pool dataset with a persistent DOI.
2026–	Neural Corrector for SPH Initial Configurations A graph neural network that maps disordered particle clouds to relaxed SPH initial configurations while enforcing minimum nearest-neighbour distances, replacing costly iterative relaxation. With F. Thiery, S. Adami and F. Dietrich.
2026–	Geometry-Aware Uncertainty for Melt-Pool Profiles A conditional VAE ensemble that predicts melt-pool geometry from process parameters with a calibrated geometric uncertainty, used to separate conduction- and keyhole-mode regimes and to drive active exploration of the process window.
2026–	Uncertainty-Aware Latent-Space Bayesian Optimisation Gaussian processes on the latent space of conditional VAEs to search process parameters while accounting for model uncertainty.

PUBLICATIONS

2024	Mueller et al. <i>InsertDiffusion: Identity Preserving Visualization of Objects through a Training-Free Diffusion Architecture</i> . arXiv:2407.10592 .
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TEACHING AND SUPERVISION

2026–	Teaching Assistant , Algorithms for Uncertainty Quantification, TUM Designed and taught tutorials on Gaussian process regression and Markov chain Monte Carlo, including notebooks and exercise sheets.
2026–	Teaching Assistant , Scientific Machine Learning 1, TUM
2025–	Student Supervision , TUM Master's theses, guided research and interdisciplinary projects on simulation data pipelines and simulator adapters, Gaussian process surrogates, active learning and uncertainty-aware conditional VAEs.
2025–	Tutor , TUM: Junge Akademie Mentoring a team of scholarship holders through an interdisciplinary research project: scoping, methodology, project management and presentation.
2021, 2022	Student Teaching Assistant , Discrete Probability Theory, TUM

INDUSTRY EXPERIENCE

2023–2024	Research Intern , BMW, Munich Generative AI for product development: diffusion models and inference-time conditioning. Contributed to <i>InsertDiffusion</i> .
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SELECTED PROJECTS

2023	TUM Data Innovation Lab , AI-Driven Design Optimization with Physics Simulators Reinforcement learning for topology optimisation: a DQN agent removes material from a 2D plate under stress to reduce volume while keeping strain energy low.
2023	Practical Course , Machine Learning in Crowd Modeling and Simulation From dimensionality reduction (PCA, manifold learning) to autoencoders for crowd modelling.

AWARDS AND ACTIVITIES

2021–2023	Scholarship Holder , TUM: Junge Akademie. Task force on external partnerships; project group designing an accessible 3D-printable finger prosthesis.
2022–2023	TUM Science Hackathon , organising staff (2022), participant (2023)
2018	HackaTUM , first prize, Ledvance challenge
2017	International Space Settlement Design Competition , first place, international final, Kennedy Space Center
2017	CanSat (ESA) , first prize, national final

SKILLS

Advanced	Python, PyTorch, NumPy/SciPy, Java
Intermediate	C, C++, SQL, Bash, Git, SLURM/HPC, Docker/Enroot, ParaView, GPyTorch/BoTorch

LANGUAGES

Romanian	Native	
English	C1	IELTS, 2017
German	C1	Deutsches Sprachdiplom II, 2017
Italian	B1	